

Shibaura Plant Feedback Management System

PRODUCTION SERVER DEPLOYMENT MANUAL — shibaura.amoebatronix.com

Domain: `http://shibaura.amoebatronix.com` | Path: `/home/john/shibaura/feedback_management` | User: `john`

Parameter	Production Specification	Role / Scope
Target Domain	<code>http://shibaura.amoebatronix.com</code> (SSL Ready)	Public Gateway
Server Path	<code>/home/john/shibaura/feedback_management</code>	Application Root Directory
Web Document Root	<code>/home/john/shibaura/feedback_management/public</code>	Nginx Web Root Directory
System User	<code>john</code> (Member of <code>www-data</code> group)	Process & File Ownership
Backend Stack	Laravel 12.x / PHP 8.2+ FPM / MySQL 8.0	Framework & Database Engine
Frontend & Auth	Vite 6 (Node 20) & JWT Auth (HS256)	UI Bundles & Mobile App Tokens

1. Server Preparation & Firewall Configuration

Connect to your server via SSH as user `john` and prepare system packages:

```
ssh john@your_server_ip
sudo apt update && sudo apt upgrade -y
sudo apt install -y git curl wget zip unzip ufw software-properties-common ca-certificates
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw --force enable
```

2. Installing PHP 8.2 & Core Extensions

Install PHP 8.2, required PHP extensions for Laravel 12 & JWT, and Composer 2:

```
sudo add-apt-repository ppa:ondrej/php -y && sudo apt update
sudo apt install -y php8.2 php8.2-fpm php8.2-cli php8.2-common php8.2-mysql \
php8.2-xml php8.2-mbstring php8.2-curl php8.2-zip \
php8.2-gd php8.2-intl php8.2-bcmath
# Install Composer 2 globally:
curl -sS https://getcomposer.org/installer | php
sudo mv composer.phar /usr/local/bin/composer
```

3. MySQL 8.0 Database Setup & Dedicated User

Install MySQL 8.0, secure the engine, and provision the dedicated database and credentials:

```
sudo apt install -y mysql-server && sudo mysql_secure_installation
# Run in MySQL shell:
sudo mysql -u root -p
CREATE DATABASE plant_feedback CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'shibaura_admin'@'localhost' IDENTIFIED BY 'StrongSecretPass@2026';
GRANT ALL PRIVILEGES ON plant_feedback.* TO 'shibaura_admin'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

4. Installing Node.js 20 LTS (Vite Compilation)

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

5. Codebase Setup in /home/john/shibaura/feedback_management

Clone the application repository from GitHub into the user path and install PHP packages:

```
mkdir -p /home/john/shibaura
cd /home/john/shibaura
git clone https://github.com/Santhoshkumarsk10/feedback_management.git feedback_management
cd /home/john/shibaura/feedback_management
git checkout develop_santhosh # or main
composer install --no-dev --optimize-autoloader --no-interaction
```

6. Production Environment (.env) & Cryptographic Keys

Configure production variables in `.env` and generate the application and JWT secrets:

```
cp .env.example .env
nano .env

# Essential .env values for shibaura.amoebatronix.com:
APP_NAME="Shibaura Plant Feedback"
APP_ENV=production
APP_DEBUG=false
APP_URL=http://shibaura.amoebatronix.com
APP_TIMEZONE=Asia/Kolkata

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=plant_feedback
DB_USERNAME=shibaura_admin
DB_PASSWORD=StrongSecretPass@2026

SESSION_DRIVER=database
CACHE_STORE=database
QUEUE_CONNECTION=database
```

```
php artisan key:generate
php artisan jwt:secret
php artisan storage:link
```

CRITICAL: Running `php artisan jwt:secret` is mandatory! The mobile application and visitor authentication API will fail with a 500 error if `JWT_SECRET` is empty.

7. Database Migrations & Initial Master Seeders

Execute database migrations followed by the project seeders to populate plants, questionnaires, and users:

```
php artisan migrate --force
php artisan db:seed --class=PlantSeeder --force
php artisan db:seed --class=MasterSeeder --force
php artisan db:seed --class=CompanyAndBannerSeeder --force
```

NOTE: Default accounts seeded: Super Admin (`superadmin@plant.test`), Admin (`admin@plant.test`), Organizers (`ravi@plant.test`). Default password: `password`. Change immediately upon login.

8. Compiling Production Frontend Assets (Vite)

Compile production CSS/JS bundles into `public/build`:

```
npm install
npm run build
```

9. Nginx Web Server Configuration for shibaura.amoebatronix.com

Configure Nginx with root `/home/john/shibaura/feedback_management/public`:

```
sudo apt install -y nginx
sudo cp /home/john/shibaura/feedback_management/deployment/nginx-feedback.conf \
/etc/nginx/sites-available/shibaura.amoebatronix.com

sudo ln -s /etc/nginx/sites-available/shibaura.amoebatronix.com /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t && sudo systemctl reload nginx
```

10. Home Directory Permissions Fix (Resolving 403 Forbidden)

When serving from `/home/john/`, Nginx (`www-data`) requires directory traversal permissions:

```
# Allow Nginx (www-data) to traverse into /home/john:
chmod +x /home/john
chmod +x /home/john/shibaura

# Add www-data user to group john:
sudo usermod -a -G john www-data

# Set ownership and permissions on writable Laravel directories:
sudo chown -R john:www-data /home/john/shibaura/feedback_management/storage
sudo chown -R john:www-data /home/john/shibaura/feedback_management/bootstrap/cache
sudo chmod -R 775 /home/john/shibaura/feedback_management/storage
sudo chmod -R 775 /home/john/shibaura/feedback_management/bootstrap/cache
chmod 600 /home/john/shibaura/feedback_management/.env
```

CRITICAL: Skipping `chmod +x /home/john` will cause Nginx to throw a **403 Forbidden** error because `www-data` cannot read the public folder.

11. Production Caching & Performance Optimization

Warm up Laravel configuration, routing, and blade view caches for peak performance:

```
cd /home/john/shibaura/feedback_management
php artisan config:cache
php artisan route:cache
php artisan view:cache
php artisan event:cache
```

12. SSL Certificate Installation (Let's Encrypt HTTPS)

Secure domain with free Let's Encrypt SSL certificate:

```
sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d shibaura.amoebatronix.com
```

13. Background Queues (Supervisor) & Cron Scheduler

A. Crontab for Laravel Scheduler:

```
crontab -e
# Add line:
* * * * * cd /home/john/shibaura/feedback_management && php artisan schedule:run >> /dev/null 2>&1
```

B. Supervisor for Queue Worker (Database Queue):

```
sudo apt install -y supervisor
sudo cp /home/john/shibaura/feedback_management/deployment/supervisor-queue.conf \
/etc/supervisor/conf.d/shibaura-worker.conf
sudo supervisorctl reread && sudo supervisorctl update
sudo supervisorctl start shibaura-feedback-worker:*
```

14. Automated Future Deployments (1-Command Update Script)

Whenever updates are pushed to GitHub, update the server with a single command:

```
bash /home/john/shibaura/feedback_management/deployment/deploy.sh
```

15. Post-Deployment Verification Checklist

- [✓] Web Admin Portal accessible at <http://shibaura.amoebatronix.com/login>
- [✓] Super Admin and Plant Admin accounts log in successfully
- [✓] Mobile App APK download link streams file at [/apk/shibaura-plant-feedback.apk](#)
- [✓] Visitor Feedback Form renders all sections, MCQs, and rating matrices
- [✓] Storage symlink verified (company logo and banner images render properly)
- [✓] Log verification: `storage/logs/laravel.log` shows zero fatal exceptions